

Hiraia — progress update

Early September 2026

An offline, on-device science tutor for Filipino grade-school students (Grades 3–10), aligned to the DepEd MATATAG curriculum, in Tagalog, Cebuano and English. The published build (v0.1) targets Android 12+ phones with 6 GB+ memory; a 1B budget tier runs on a Redmi (Snapdragon 685) in unpublished builds.

36,487

illustrated fact cards

50,279

verified trilingual facts

324

MATATAG competencies
tracked (321 with cards)

25,746

cards with a quiz question

Evidence: repository at commit 58d358495 (branch question-cards) and its evaluation records; the published app is app v0.1 — the feed shown here ships in the next build. Sources are in the slide notes.



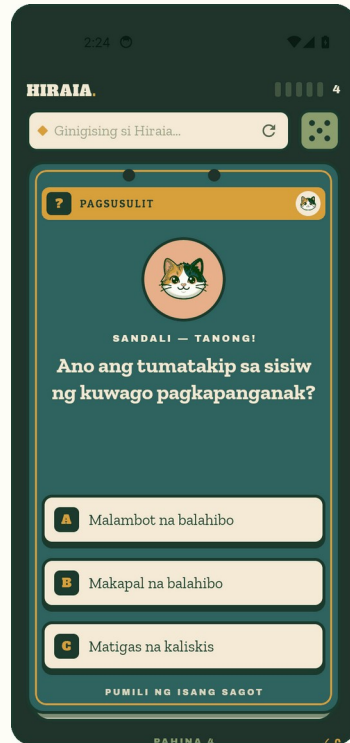
current build (card-ui, 26 Aug 2026)

hi User experience

The app teaches with a deck of flash cards drawn from the DepEd MATATAG curriculum: a child reads a card, is quizzed on what they just read, is told what they have covered, and can ask for a topic at any time.



A card carries one fact, one illustration, and the topic that comes next.



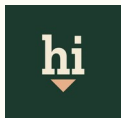
Every few cards a question interrupts the deck, on a fact just read.



A periodic recap names the topics the child actually covered.



Typing a word moves the deck to a matching card — here, "bulkan".



Training runs on RunPod

Most of the compute went into a continued-pretrained Filipino base model; the rest into tutoring fine-tunes, a pruning experiment, and evaluation.

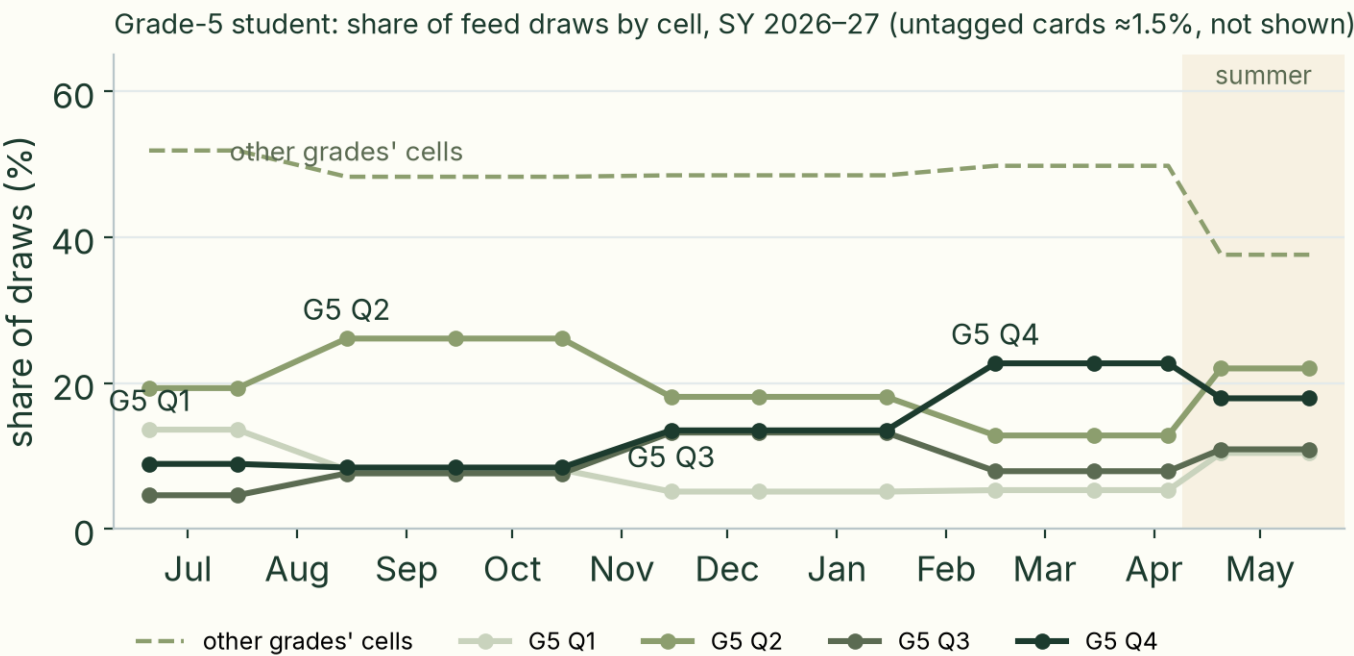
run	GPUs	GPU-h	cost	basis	outcome
Full continued pre-training — Qwen3.5-2B, ≈13.7B tokens	8 × H100 80GB	294	\$972	cost documented; hours from pod age	Held-out perplexity: Tagalog −80%, Cebuano −82%, English −4% (improved)
CPT probe run 2 (full-parameter, ZeRO-2)	8 × H100	167	\$550	documented	Only checkpoint-125 (525M tokens) survived a cross-DC sync failure — it validated the thesis: perplexity −63% tl / −63% ceb; GGUF export de-risked
CPT probe run 1 (≈5.1B tokens, trained as LoRA by mistake)	8 × H100	85	\$285	documented	No usable artefact; the pod/volume/heartbeat pipeline was proven
Full CPT attempt 1 (aborted)	8 × H100	66	\$216	estimate	Pre-processing collapse (worker count vs cgroup quota); run held and restarted
Prune + heal 2.4B experiment (Sailor2-3B)	8 × A100	56	\$86	documented	Heal recovered fluency (ppl tl 6.7 / ceb 9.9); conservative re-SFT under-fit — not shipped
GRPO rounds 1–5 (1B/3B)	1–4 × H100 / RTX 6000	55	\$74	mostly documented	Not shippable: capability 3.30 → 2.98; English +1.4 but abstention and Bisaya regressed
SFT v1 / v2 on the CPT base (Qwen3.5-2B)	1 × A100 80GB	3	\$6	documented	v1 is the ship candidate: reply-language accuracy tl 99% · ceb 99% · en 94%
Adapter iterations on Sailor2-3B / 1B and tooling (36 entries, May–June)	H100 / L40 / A100 / L4	60	\$128	mostly estimated	Shipped adapters v3 → v11: grounding faithfulness, static-prompt TTFT, kitten/cat tiers; QVAC-native training path evaluated and dropped
Checkpoint-sync, eval and debug pods	various	68	\$85	mixed	Evaluation on a device-equivalent GGUF; pre-flight debugging of run 2

≈853 GPU-hours and ≈\$2,402 on RunPod since late May 2026 (rows sum to the totals); ≈\$2,000 of that from figures recorded at the time, the rest estimated at list price — no billing export yet. About 80% of the compute went into continued pre-training of the Filipino base model.



A curriculum-aware feed

Cards are weighted by the student's grade and the curriculum quarter inferred from the device date; the boost moves through the school year by itself, and flattens across the whole grade once school is out.



fact card



interject quiz card

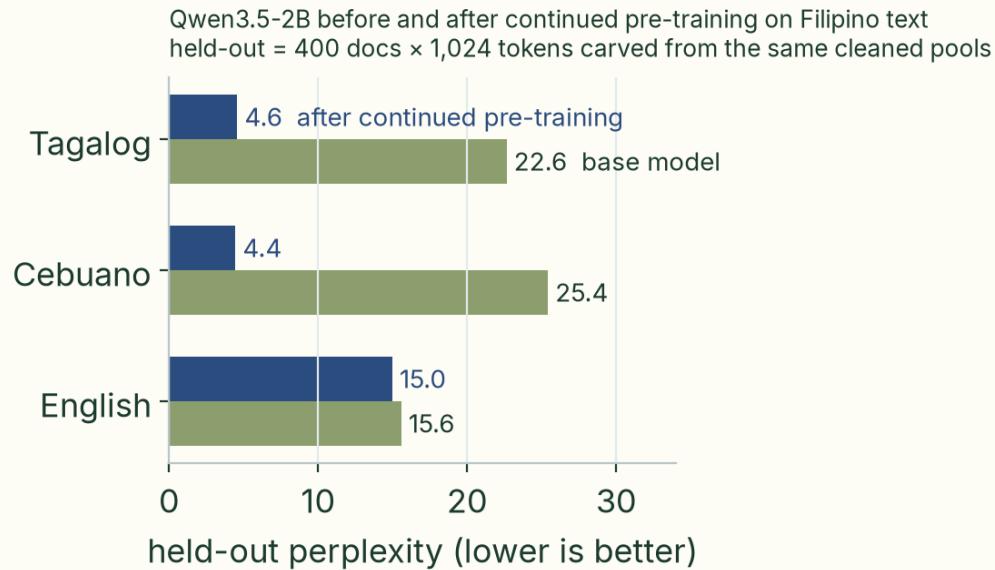
current quarter, same grade	adjacent quarter	other quarter, same grade	adjacent grade, current	school out: any cell of the grade	off-curriculum
×6	×3	×1.5	×2	×6	×0.4

- 36,487 illustrated cards; 34,719 facts labelled to 324 competencies (LLM labels, 83% inter-model agreement; single-model cells down-weighted)
- 25,746 cards (70%) carry a verified quiz question
- 0 elementary competencies below floor; 38 junior-high competencies below floor in the card pool (16 once DepEd module cards are counted)



Model progress: fluency and pedagogy

From a fluent-but-cautious 3B adapter line to a continued-pretrained 2B base that reads and writes Tagalog and Cebuano: fluency is measured on held-out text, pedagogy on scripted tutoring probes.



Tiles: pedagogy figures are from the June 3B line; language accuracy and hedging from the August 2B (SFT v1). The 2B capability run is pending.

3.49 → 3.72

capability score (0–5), 102 probes — June 3B line, rebalance only

99 · 96 · 94%

reply-language accuracy, SFT v1 2B (tl · ceb · en); Cebuano 99% corrected

64%

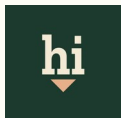
of Tagalog replies open with a hedge; two-thirds still explain the term

Pedagogy (measured on scripted probes)

Scripted probes (102 in June, 143 now; 11 tiers, LLM-judged 0–5) showed the June model fluent but over-cautious: 19 of 91 answerable questions were refused. A retrieval fix plus a small data rebalance (+82 rows) lifted the score 3.32 → 3.72 (rebalance alone +0.22), and grounding faithfulness was fixed — the tutor defers to retrieved verified facts and abstains when none apply. The 2B line inherits the same harness; its capability run is the next measurement.

Known weaknesses (measured)

- Hedging: the SFT v1 2B prefixes 64% of Tagalog replies with a 'not sure' clause on grade-5 terms; about two-thirds of those then explain the term — a template-wording effect, not a knowledge gap
- English mode-lock is the 2B's fragile edge (94% vs 99% in Tagalog and Cebuano); misroutes fall to Tagalog
- The 1B budget tier cannot learn safety negation (DPO tested and rejected); junior-high content is thinner than elementary

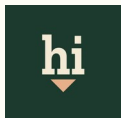


Next 2–3 weeks

Consolidate the model, then prove it on the device.

- More SFT and knowledge distillation on the CPT base (mode-locked buckets, teacher-generated tutoring turns)
- Extensive synthetic testing on-device: scripted student sessions on the Redmi, TTFT and per-page latency budgets
- Model-as-judge quality control on every generated fact, quiz and translation (decorrelated judge, AUP-safe routing)
- APK-level optimisations: bundle size (engravings subset, indexed PNG), cold start, SQLite migrations, edge-walk cost
- Native-speaker review of the Tagalog and Cebuano copy (onboarding, settings, hedging clause)
- Junior-high gap fill: Lane B facts for the 16 remaining thin competencies; module-card supply
- Finish the quiz lane (10.7k pool cards still without a question) and per-competency Miss-Card labels
- Over-the-air fact-bank updates so content ships without an APK
- Retrieval-QA cases per new competency before each gate run
- Draft a classroom-pilot protocol (consent, offline logging, pre/post measures) to propose to DepEd schools

To report next time: the 2B capability run on the same 143 probes; on-device TTFT and per-page latency on the Redmi; quiz coverage after the lane completes; bundle size after optimisation.



Links

Code, build and evaluation records are public; model weights follow once the release checklist is done.

Website

<https://hiraia.org>

APK (direct download)

<https://hiraia.org/models/hiraia.apk>

v0.1.0, 325 MB, sha256 708ea605..., Android 12+, 6 GB+ RAM; mirror:
github.com/helloluis/hiraia/releases/tag/v0.1.0

GitHub

<https://github.com/helloluis/hiraia>

source, pipelines and evaluation harness (public)

Hugging Face

to be published

model weights + evaluation kit: release pending; corpus provenance will be documented, DepEd-derived text is not redistributed